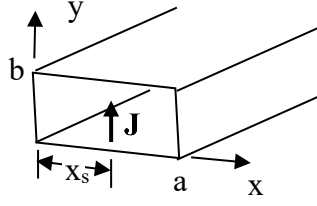
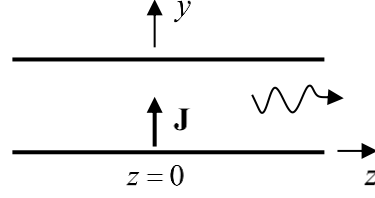


## Notes 11

### Current Sources in a waveguide



$$\mathbf{J} = I(y)\delta(x - x_s)\delta(z)\hat{\mathbf{y}}$$



cross-sectional view

$$J_y \Rightarrow A_y$$

Physical reasoning  $\Rightarrow TM_y$  modes

$$\mathbf{A} = \psi \hat{\mathbf{y}}$$

$$\mathbf{H} = \frac{1}{\mu} \nabla \times \mathbf{A} \qquad \mathbf{E} = \frac{-j\omega}{k^2} (k^2 \mathbf{A} + \nabla \nabla \cdot \mathbf{A})$$

$$H_x = -\frac{1}{\mu} \frac{\partial \psi}{\partial z} \qquad E_x = \frac{-j\omega}{k^2} \frac{\partial^2 \psi}{\partial x \partial y}$$

$$H_y = 0 \qquad E_y = \frac{-j\omega}{k^2} \left( k^2 \psi + \frac{\partial^2 \psi}{\partial y^2} \right)$$

$$H_z = \frac{1}{\mu} \frac{\partial \psi}{\partial x} \qquad E_z = \frac{-j\omega}{k^2} \frac{\partial^2 \psi}{\partial z \partial y}$$

Subject to B.C.'s on  $\mathbf{E}$  and  $\mathbf{H}$

$$\psi_{mn}^{\pm} = B_{mn}^{\pm} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{\mp jk_{z,mn}z}; \quad k_{z,mn} = \sqrt{k^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

i.e.

$$\psi^+ = \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} B_{mn}^+ \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-jk_{z,mn}z}, \quad z > 0 \quad (1)$$

$$\psi^- = \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} B_{mn}^- \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{jk_{z,mn}z}, \quad z < 0 \quad (2)$$

↗  
try combinations of  $m = 0$  and  $n = 0$

There are 2 boundary conditions at  $z = 0$  which aid in finding  $B_{mn}^{\pm}$

$$\hat{\mathbf{z}} \times (\mathbf{E}^+ - \mathbf{E}^-)_{z=0} = 0 \Rightarrow \left( E_x^+ = E_x^- \right)_{z=0} \text{ and } \left( E_y^+ = E_y^- \right)_{z=0} \quad (3)$$

and

$$\hat{\mathbf{z}} \times (\mathbf{H}^+ - \mathbf{H}^-)_{z=0} = J_y^s \hat{\mathbf{y}} \quad (4)$$

where  $J_y^s = I\delta(x - x_s)$  (notice that  $\delta(z)$  is automatically removed when  $z = 0$  in the BC)

With BC (3)

$$\begin{aligned}
 E_y^+ &= E_y^- \Big|_{z=0} \\
 \left\{ \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} B_{mn}^+ \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-jk_{z,mn}z} \left[ k^2 - \left(\frac{n\pi}{b}\right)^2 \right] \right. \\
 &= \left. \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} B_{mn}^- \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{jk_{z,mn}z} \left[ k^2 - \left(\frac{n\pi}{b}\right)^2 \right] \right\}_{z=0}
 \end{aligned}$$

Apply orthogonality by multiplying both sides by  $\sin\left(\frac{m'\pi}{a}x\right)\cos\left(\frac{n'\pi}{b}y\right)$  and integrate over the waveguide cross-section

$$\Rightarrow B_{mn}^+ = B_{mn}^- \doteq B_{mn}$$

Now apply boundary condition (4)

$$\begin{aligned}
 \hat{\mathbf{n}} \times (\mathbf{H}^+ - \mathbf{H}^-)_{z=0} &= J_y^s \hat{\mathbf{y}} \\
 \underbrace{(\hat{\mathbf{z}} \times \hat{\mathbf{x}})}_{\hat{\mathbf{y}}} (H_x^+ - H_x^-)_{z=0} &= J_y^s \hat{\mathbf{y}}
 \end{aligned} \tag{4}$$

Expanding

$$\begin{aligned}
 \frac{1}{\mu} \left\{ \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} B_{mn} \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) (+jk_{z,mn}) e^{-jk_{z,mn}z} \right. \\
 \left. - \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} B_{mn} \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) (-jk_{z,mn}) e^{jk_{z,mn}z} \right\}_{z=0} = J_y^s
 \end{aligned}$$

or

$$\frac{1}{\mu} \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} 2B_{mn} (jk_{z,mn}) \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) = J_y^s$$

Again by orthogonality – multiply by  $\sin\left(\frac{m'\pi}{a}x\right)\cos\left(\frac{n'\pi}{b}y\right)$  and integrate over waveguide cross-section – result  $\rightarrow$

$$B_{mn} = \frac{-j\mu}{k_{z,mn}} \left(\frac{\varepsilon_n}{ab}\right) \int_{x=0}^b \int_{y=0}^a J_y^s \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) dx dy \tag{5}$$

$$\varepsilon_n = \begin{cases} 1; & n = 0 \\ 2; & n > 0 \end{cases} \quad (\text{Neumann's number})$$

$B_{mn}$  can be substituted back into (1) and (2) and the fields are therefore completely determined in terms of the current  $J_y^s$

Note that:

$$J_y^s = \delta(x - x_s) I(y)$$

so

$$\begin{aligned} B_{mn} &= \frac{-j\mu}{k_{z,mn}} \left( \frac{\varepsilon_n}{ab} \right) \int_{y=0}^b \int_{x=0}^a I(y) \delta(x - x_s) \sin\left(\frac{m\pi}{a} x\right) \cos\left(\frac{n\pi}{b} y\right) dx dy \\ &= \frac{-j\mu}{k_{z,mn}} \left( \frac{\varepsilon_n}{ab} \right) \sin\left(\frac{m\pi}{a} x_s\right) \int_{y=0}^{y_s} I(y) \cos\left(\frac{n\pi}{b} y\right) dy \quad (6) \end{aligned}$$

From (1)

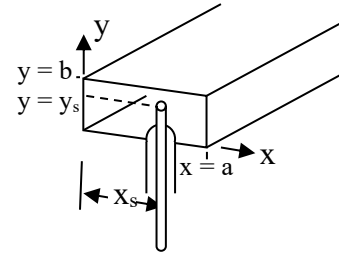
$$\psi^+ = \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} B_{mn} \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-jk_{z,mn}z}, \quad z > 0$$

Inserting  $B_{mn}$  from above

$$\psi^+ = \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} \frac{-j\mu}{k_{z,mn}} \left( \frac{\varepsilon_n}{ab} \right) \sin\left(\frac{m\pi}{a} x_s\right) \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) \int_{y=0}^{y_s} I(y) \cos\left(\frac{n\pi}{b} y\right) dy e^{-jk_{z,mn}z}, \quad z > 0 \quad (7)$$

so for example:

$$\begin{aligned} H_x &= -\frac{1}{\mu} \frac{\partial \psi}{\partial z} \\ &= \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} \left( \frac{\varepsilon_n}{ab} \right) \sin\left(\frac{m\pi}{a} x_s\right) \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) \int_{y=0}^{y_s} I(y) \cos\left(\frac{n\pi}{b} y\right) dy e^{-jk_{z,mn}z}, \quad z > 0 \quad (8) \end{aligned}$$



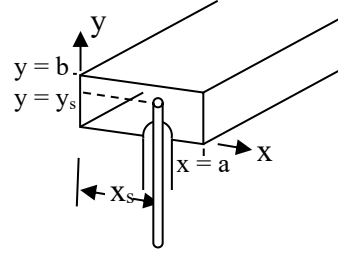
## Special Dominant Mode Case for a Source in a Waveguide

(1) Assume that the waveguide is operating in its lowest order mode,  $m = 1, n = 0$ . This gives

$$\psi^+ = B_{10} \sin\left(\frac{\pi}{a}x\right) e^{-jk_{z,10}z} \quad (9)$$

$$k_{z,10} = \sqrt{k^2 - \left(\frac{\pi}{a}\right)^2}$$

$$B_{10} = \frac{-j\mu}{k_{z,10}} \left(\frac{1}{ab}\right) \sin\left(\frac{\pi}{a}x_s\right) \int_{y=0}^{y_s} I(y) dy \quad (10)$$



From page 12.1 and  $\psi^+$  above, the  $TM_{y,10}$  mode fields are

$$\left. \begin{aligned} H_x &= \frac{jk_{z,10}}{\mu} B_{10} \sin\left(\frac{\pi}{a}x\right) e^{-jk_{z,10}z} \\ H_y &= 0 \\ H_z &= \frac{\pi}{\mu a} B_{10} \cos\left(\frac{\pi}{a}x\right) e^{-jk_{z,10}z} \\ E_x &= 0 \\ E_y &= -j\omega B_{10} \sin\left(\frac{\pi}{a}x\right) e^{-jk_{z,10}z} \\ E_z &= 0 \end{aligned} \right\} \Rightarrow \text{which is also } TE_{z,10} \quad (11)$$

Usually

$$I(y) \doteq I_0 \sin[k(y_s - y)] \quad (12)$$

is a good approximation to the probe current. A frequency where  $y_s = \lambda/4$  is ideal because the probe length with its image  $= \lambda/2$ . Also the probe is placed at  $x_s = a/2$ , because the field  $E_y$  is a maximum at that point.

The magnetic field in the coax is  $\mathbf{H} = \frac{V}{\eta \rho \ln(b'/a')} e^{-jkz} \hat{\phi}$  where  $a'$  and  $b'$  are the center and outer conductor radii,  $\rho$  is the radial cylindrical coordinate and  $V$  is the coaxial line generator voltage. Assuming the coax is impedance matched to the rectangular waveguide, the coaxial aperture magnetic field is therefore  $\mathbf{H} \cong \frac{V}{\eta \rho \ln(b'/a')} \hat{\phi}$ . Since the field in a coaxial cable is a TEM mode, the electric current amplitude  $I_0$  on the wire is determined by

$$I_0 = \oint \mathbf{H} \cdot d\mathbf{l} \text{ so } I_0 \cong \int_0^{2\pi} \left( \frac{V}{\eta \rho \ln(b'/a')} \hat{\phi} \right) \cdot \rho d\phi \hat{\phi} = \frac{2\pi V}{\eta \ln(b'/a')}. \quad (13)$$

Substituting (10), (12) and (13) into (11) gives an analytical solution the  $TE_{10}$  mode fields in terms of the source voltage  $V$ .

2) In the case above  $a$  and  $b$  are chosen so that at the operation frequency

$$k^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2 \begin{cases} > 0 & \text{when } m=1, n=0 \\ < 0 & \text{when } m \text{ and } n \text{ are any integer combination} \\ & \text{except } m=1, n=0 \end{cases} \quad (14)$$

$$\text{for } \begin{cases} m=1, n=0 & k_{z,10} = \sqrt{k^2 - \left(\frac{\pi}{a}\right)^2} \quad \text{which is real} \\ m=1, 2, \dots; n=0, 1, 2, \dots \\ \quad (\text{except } m=1, n=0) & k_{z,mn} = -j\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - k^2} \\ & = -j\alpha_{mn} \end{cases} \quad (15)$$

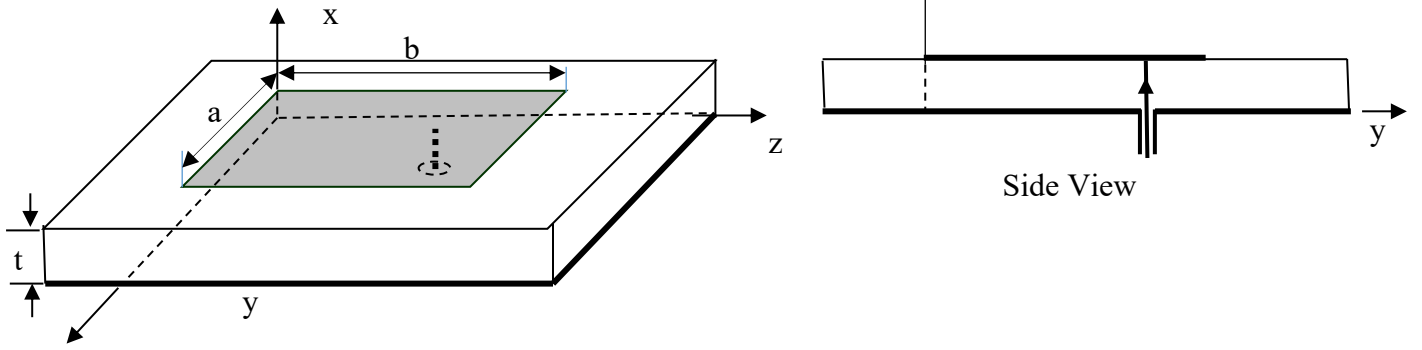
So the complete electric field becomes

$$E_y = -j\omega B_{10} \sin\left(\frac{\pi}{a}x\right) e^{-jk_{z,10}z} - \frac{j\omega}{k^2} \sum_{\substack{m=1 \\ \text{except } 1,0}}^{\infty} \sum_{n=0}^{\infty} B_{mn} \left( k^2 - \left(\frac{n\pi}{b}\right)^2 \right) \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-\alpha_{mn}z} \quad (16)$$

The first term is a propagating wave. The second term contains the evanescent modes (or non-propagating modes) which account for energy storage around the source. As the frequency goes up successive  $k_{z,mn}$  become real and these modes begin to propagate.

Also see  
Harrington Chapt. 8  
and  
Collin Field Theory of Guided Waves  
latest edition

## Microstrip Patch Antenna



Assume a constant amplitude source at  $x_s, z_s$  extends up through the thickness  $t$  of the waveguide,  $\mathbf{J} = I(x)\delta(y - y_s)\delta(z - z_s)\hat{\mathbf{x}}$

$$\mathbf{A} = \psi \hat{\mathbf{x}} \Rightarrow \mathbf{H} = \frac{1}{\mu} \nabla \times \psi \hat{\mathbf{x}}, \quad \mathbf{E} = -\frac{j\omega}{k^2} \left( k^2 \psi \hat{\mathbf{x}} + \nabla \frac{\partial}{\partial x} \psi \right)$$

The source direction  $J_x$  suggests solve  $TM_x$  from which we get

$$\begin{aligned} H_x &= 0 & E_x &= -\frac{j\omega}{k^2} \left( k^2 \psi + \frac{\partial^2}{\partial x^2} \psi \right) \\ H_y &= \frac{1}{\mu} \frac{\partial}{\partial z} \psi & E_y &= \frac{-j\omega}{k^2} \frac{\partial^2}{\partial y \partial x} \psi \\ H_z &= -\frac{1}{\mu} \frac{\partial}{\partial y} \psi & E_z &= \frac{-j\omega}{k^2} \frac{\partial^2}{\partial z \partial x} \psi \end{aligned}$$

First set  $\mathbf{J} = 0$  to obtain the ‘natural response’ of the microstrip resonator, which are the solutions to

$$\nabla_t^2 \psi + k^2 \psi = 0 \quad \text{where} \quad \nabla_t^2 = \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$H = 0 \text{ tangential to the boundaries} \Rightarrow \psi = A \cos k_y y \cos k_z z$$

$$H_z(y=a) = 0 \Rightarrow \sin(k_y a) = 0$$

$$\begin{aligned} \text{B.C. on } H_z : & \\ \Rightarrow k_y &= \frac{m\pi}{a}, \quad m = 0, 1, 2, 3, \dots \end{aligned}$$

$$H_y(z=b) = 0 \Rightarrow \sin(k_z b) = 0$$

$$\begin{aligned} \text{B.C. on } H_y : & \\ \Rightarrow k_z &= \frac{n\pi}{b}, \quad n = 0, 1, 2, 3, \dots \end{aligned}$$

$$\therefore \psi = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} A_{mn} \cos(k_{y,m} y) \cos(k_{z,n} z) \quad (1)$$

From Helmholtz equation, Notes page 1-6

$$\nabla^2 \mathbf{E} + k^2 \mathbf{E} = j\omega\mu \mathbf{J} - \frac{\nabla(\nabla \cdot \mathbf{J})}{\sigma + j\omega\epsilon} + \nabla \times \mathbf{M}$$

with  $J_x$  constant,  $J_y = J_z = 0 \Rightarrow (\nabla \cdot \mathbf{J} = 0)$  and  $\mathbf{M} = 0$  :

$$\begin{aligned} \nabla^2 E_x + k^2 E_x &= j\omega\mu J_x \\ \nabla_t^2 E_x + \underbrace{\frac{\partial^2}{\partial x^2} E_x}_{=0 \text{ no } x\text{-variation}} + k^2 E_x &= j\omega\mu J_x \end{aligned} \quad (2)$$

Also

$$E_x = -\frac{j\omega}{k^2} \left( k^2 \psi + \frac{\partial^2}{\partial x^2} \psi \right) = -j\omega\psi \quad (3)$$

Substitute (1) and (3) into (2)

$$\begin{aligned} -j\omega(\nabla_t^2 + k^2) \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} A_{mn} \cos k_y y \cos k_z z &= j\omega\mu J_x \\ -\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} A_{mn} \underbrace{(-k_y^2 - k_z^2 + k^2)}_{\doteq -k_{mn}^2} \cos k_y y \cos k_z z &= \mu I \delta(y - y_s) \delta(z - z_s) \end{aligned} \quad (4)$$

From orthogonality

$$\int_0^b \int_0^a \cos k_y y \cos k_z z \cos k'_y y \cos k'_z z dy dz = \begin{cases} \frac{ab}{\epsilon_m \epsilon_n} & \epsilon_m = 2 \text{ if } m = m', \epsilon_m = 1 \text{ if } m = m' = 0 \\ & \epsilon_n = 2 \text{ if } n = n', \epsilon_n = 1 \text{ if } n = n' = 0 \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

and

$$\int_0^b \int_0^a \delta(y - y_s) \delta(z - z_s) \cos k'_y y \cos k'_z z dy dz = \cos k'_y y_s \cos k'_z z_s \quad (6)$$

These results in (4) gives

$$\begin{aligned} A_{mn} &= \frac{-\epsilon_m \epsilon_n \mu I}{ab(k^2 - k_{mn}^2)} \cos k_y y_s \cos k_z z_s \\ \therefore E_x(y, z) &= \frac{jk\eta I}{ab} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\epsilon_m \epsilon_n}{(k^2 - k_{mn}^2)} \cos k_y y \cos k_z z \cos k_y y_s \cos k_z z_s \end{aligned} \quad (7)$$

$$\varepsilon_{m,n} \doteq \text{Neumann Number} = \begin{cases} 1 & \text{if } n,m=0 \\ 2 & \text{if } n,m \neq 0 \end{cases}$$

This method gives all of the modes that exist in the microstrip antenna cavity. However, the mode  $k_{MN}$  where  $k_{MN}^2 \simeq k^2$  is the dominant mode because this mode has the largest amplitude due to the almost singular coefficient.

$$\begin{aligned} E_x(y, z) &= \frac{jk\eta I}{ab} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\varepsilon_m \varepsilon_n}{(k^2 - k_{mn}^2)} \cos k_y y \cos k_z z \cos k_y y_s \cos k_z z_s \\ &\cong \frac{jk\eta I}{ab} \frac{\varepsilon_M \varepsilon_N}{(k^2 - k_{MN}^2)} \cos k_y y \cos k_z z \cos k_y y_s \cos k_z z_s + \text{higher order modes} \end{aligned} \quad (8)$$

We usually design the microstrip antenna so that the mode is 1,0 or 0,1. Suppose that it is the  $M = 1, N = 0$  mode. Then

$$E_x(y) \cong \frac{jk\eta I}{ab} \frac{2}{\left[ k^2 - \left( \frac{\pi}{a} \right)^2 \right]} \cos \left( \frac{\pi y}{a} \right) \cos \left( \frac{\pi y_s}{a} \right) \quad (9)$$

The wavenumber  $k$  is determined by the given frequency and substrate material. The dimension  $a$  is found by setting  $k^2 - \left( \frac{\pi}{a} \right)^2 = 0$  which leads to  $a = \lambda / 2$ , the resonant length.

When  $k = \pi/a$  the 1,0 mode amplitude is not actually infinity because in actual practice  $k$  contains loss.

The magnetic currents  $\mathbf{M}_s = \mathbf{E} \times \hat{\mathbf{n}}$  at  $y = 0$  and  $y = a$  are determined to be

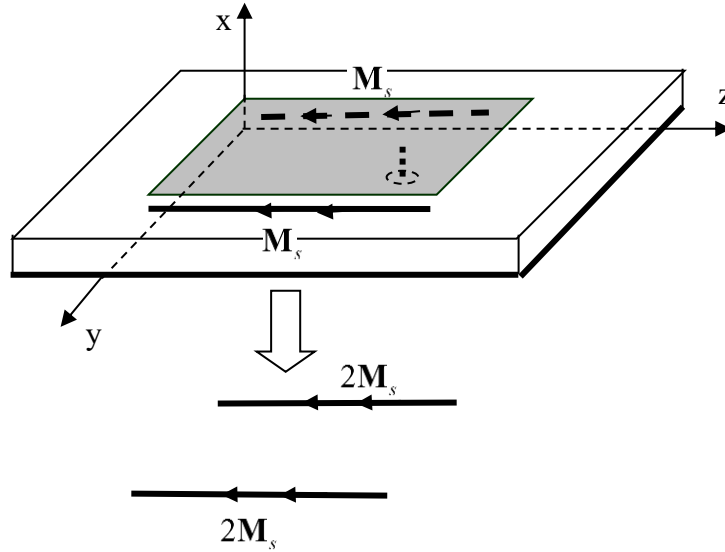
$$\mathbf{M}_s(y=0) = E_x(y=0, z) \hat{\mathbf{x}} \times -\hat{\mathbf{y}} = E_x(y=0, z)(-\hat{\mathbf{z}}) = -\frac{jk\eta I}{ab} \frac{2}{\left[ k^2 - \left( \frac{\pi}{a} \right)^2 \right]} \cos \frac{\pi y_s}{a} \hat{\mathbf{z}} \quad (10)$$

$$\mathbf{M}_s(y=a) = E_x(y=a, z) \hat{\mathbf{x}} \times \hat{\mathbf{y}} = E_y(y=a, z) \hat{\mathbf{z}} = -\frac{jk\eta I}{ab} \frac{2}{\left[ k^2 - \left( \frac{\pi}{a} \right)^2 \right]} \cos \frac{\pi y_s}{a} \hat{\mathbf{z}} \quad (11)$$

which have the same amplitude and are in the same direction. Notice that the position of the feed current source does not greatly affect the modes that exist in the cavity, but it does affect the antenna impedance seen by the coax feed line. The position of the feed source can therefore be moved to a place where the antenna input impedance matches the



coax feed line characteristic impedance. Typically, we ignore the dielectric and just image the current in the ground plane as shown at the bottom of the figure below. Since the currents are in phase they create a main beam maximum directly above the microstrip antenna. The equivalent current  $\mathbf{J}_s = \hat{\mathbf{n}} \times \mathbf{H} \cong 0$  at  $y = 0$  and  $y = a$  because at an open end of a waveguide  $\mathbf{H}_{\text{tang}} \cong 0$  while  $\mathbf{E}_{\text{tang}}$ , which follows the behavior of transmission line voltage, is a maximum at an open end.



When calculating the far fields, the coordinate system is shifted over to the center of the two magnetic currents which does not change their amplitude, phase, or field distribution. Notice the two current elements are pointed in the  $z$ -direction, a nice advantage because the radiated far fields can be calculated from

$$\begin{aligned} H_\theta &= j\omega F_z \cos\theta \\ E_\phi &= \eta H_\theta \end{aligned} \quad (12)$$